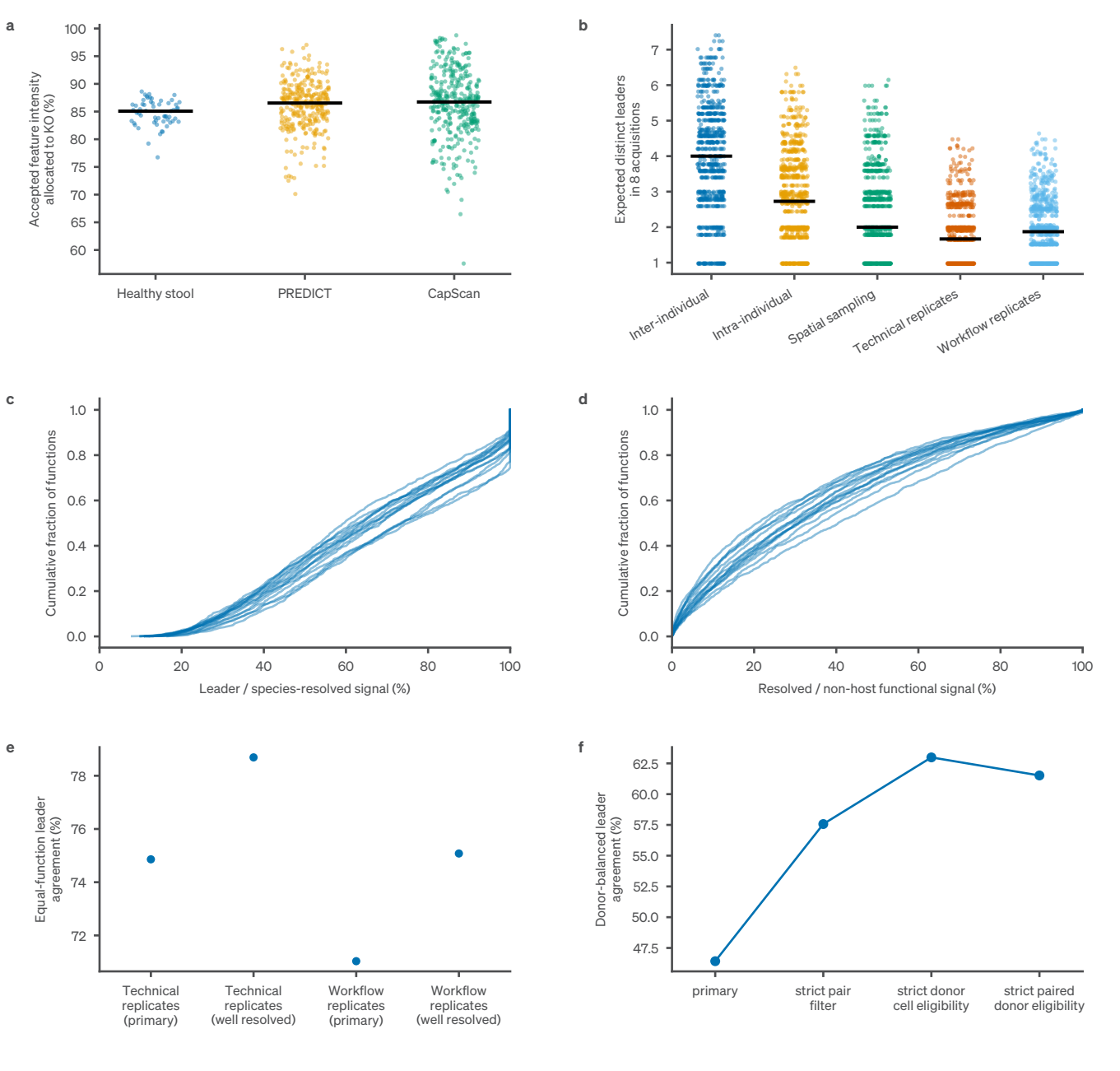

a
Accepted feature intensity allocated to KO (%)
100
95
90
85
80
75
70
65
60
Healthy stool
PREDICT
CapScan
b
Expected distinct leaders in 8 acquisitions
7
6
5
4
3
2
1
Inter-individual
Intra-individual
Spatial sampling
Technical replicates
Workflow replicates
c
Cumulative fraction of functions
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Leader / species-resolved signal (%)
d
Cumulative fraction of functions
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Resolved / non-host functional signal (%)
e
Equal-function leader agreement (%)
78
76
74
72
Technical replicates (primary)
Technical replicates (well resolved)
Workflow replicates (primary)
Workflow replicates (well resolved)
f
Donor-balanced leader agreement (%)
62.5
60.0
57.5
55.0
52.5
50.0
47.5
primary
strict pair filter
strict donor cell eligibility
strict paired donor eligibility